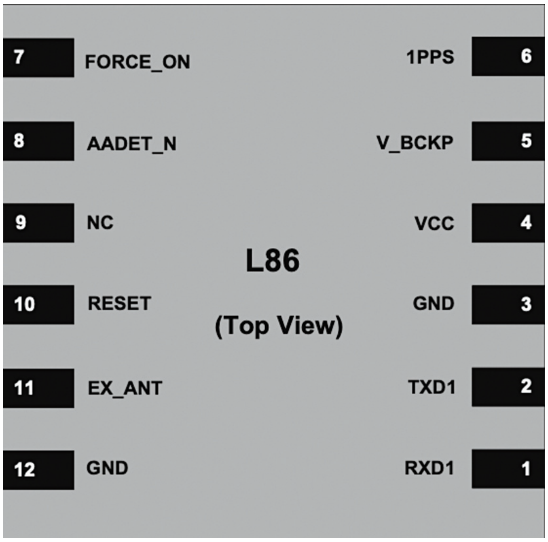

Fig. 4: GPS SMD pinouts (Source: Datasheet of L86 GPS module)

tached without taking up much space. Here, a small-sized GPS tracker is described, which can be used as a keyring or attached as a tag to children's bags to track them, or attached to the neck of animals, vehicles, parcels, or any person for tracking.

As the main objective is to keep the design size small, the IndusBoard Coin is used. Additionally, to keep the GPS size small, the L86 GPS module was chosen because of its small size with an attached GPS antenna. Next, the SIM800l module is used for sending data using GSM in its 2G band-based module. However, one can replace it with any other 4G/5G GSM module. Fig. 1 shows the author's prototype. To make this tiny GPS tracker, the components required are shown in Bill of Materials table.

BILL OF MATERIALS

Components	Description	Quantity
IndusBoard	3cm sized development board	1
L86 GPS module	GPS module	1
SIM800l module	GSM module	1
SSD1306 OLED	OLED	Optional

```
#include <HardwareSerial.h>
#include "TinyGPS++.h"
#include "AsyncSMS.h"
#include "SIM.h" // Include library for SIM module, change the library name as needed

// Define Serial port for GPS
HardwareSerial gpsSerial(2); // Pins 43 (RX) and 44 (TX)

AsyncSMS smsHelper(&Serial, 4800);

TinyGPSPlus gps;

unsigned long lastSendTime = 0;
const unsigned long sendInterval = 60000; // Send location every 60 seconds

void setup() {
  Serial.begin(115200);
  gpsSerial.begin(9600, SERIAL_8N1, 43, 44);

  // Initialize SIM module
  SIM.begin(); // Replace with appropriate initialization function for your SIM module

  smsHelper.init();
  smsHelper.smsReceived = &messageReceived;
}

void loop() {
  // Check GSM network connection status
  if (SIM.isNetworkConnected()) {
    Serial.println("GSM module connected to network.");
  } else {
    Serial.println("GSM module not connected to network.");
  }

  while (gpsSerial.available() > 0) {
    if (gps.encode(gpsSerial.read())) {
      if (gps.location.isValid() && gps.location.age() < 2000) { // Ensure GPS data is valid and recent
        unsigned long currentMillis = millis();
        if (currentMillis - lastSendTime >= sendInterval) {
          sendLocationSMS(gps.location.lat(), gps.location.lng());
          lastSendTime = currentMillis;
        }
      }
    }
```

Fig. 5: Snippet of the source code

Circuit and working

The circuit diagram of the smallest GPS tracker is shown in Fig. 2. It is built around IndusBoard, L86 GPS module, and SIM800l module. The SSD1306 OLED is also connected in the circuit, which is optional. If the display is not desired, it can be removed from the circuit.

In the design, the L86 GPS module uses the serial port with a baud rate of 9600 to send GPS data. The IndusBoard supports both software and hardware. The L86 GPS pins RXD1 and TXD1 are connected to TX and RX1 of IndusBoard, respectively.